Bayesian Analysis Tutorial Hand-in 1

Instructions

- The following tutorial counts 7.5% towards your final mark of the course.
- Submit your tutorial on Amathuba by 10 am on 24 August 2026.
- Hand in both an R markdown file and a compiled pdf version of your R markdown file. Your pdf should not contain any code in the main part of your submission. If you need to describe an algorithm this should be done in text or in the form of pseudo-code.
- Your document should contain enough information, description of methods, interpretation of results and conclusions to explain clearly what you have done. It should be clear what you have done without having to refer to any code. This means that you need to specify all settings used, even if these are default settings. A good guideline is to give enough detail to allow another person to replicate what you have done exactly, even if they want to replicate this using a different software package.
- The page limit for the main section of your hand in is 3 pages - which excludes your code! (hard constraint)
- Please include your R code in your pdf as an appendix. Upload a separate R file as well so that I can check your code where needed.
- You are allowed to consult with each other but your should hand in your own code and your own report.

Use the `SAheart` data set from the `ElemStatLearn` R package in order to answer the questions below. The package is no longer on CRAN but you can download it from `https://cran.r-project.org/src/contrib/Archive/ElemStatLearn/`. Download one of the 2019 archived packages or use either SAheart.csv or SAheart.xlsx to import the data set. The data is a subset of the data described in Rousseauw et al, (1983).

The data set contains a number of variables but for this tutorial consider the logistic regression model that include the following explanatory variables: `age`, `famhist` and `tobacco`. The response variable, `chd` is a binary variable and is coded `1` if the patient has coronary heart disease and `0` otherwise.

We wish to fit a Bayesian logistic regression model and thus require the joint prior distribution of the regression parameters. Two experts were consulted in order to elicit the prior distribution of the regression parameters. The first expert, when presented with the design points in Table 1, provided the median probability that a patient would develop coronary heart disease, `Expert1_Median`, while the second expert provided the values of the lower and upper quartile values associated with the probability that a patient would develop coronary heart disease and are denoted as `Expert2_25%` and `Expert2_75%` respectively.

Use the training data set using the R code below to answer all the questions below.

```
#you might have to first install the package (see above)
require(ElemStatLearn)
data("SAheart")

#set a random training data set
set.seed(1)
training <- SAheart[sample(1:462, size=300), ]
```

(a) Use the information obtained from Expert 1 to obtain an elicited prior distribution for the regression parameters of the logistic regression model. Use the methods proposed by Choy et al (2009).

(b) Now use the information obtained from Expert 2 to perform the same task. Use the method proposed by Hosack et al (2017). When estimating m_i and v_i, use the Kullback Leibler divergence and not the square error loss function.

(c) Contrast the **two** prior distributions obtained. i.e. compare the results obtained when using (a) and (b).

Do not use the `indirect` R package at all! You should code the 'Hosack method' yourself but you are allowed to use any R package to perform Beta regression. Explain your workings and your results clearly.

Table 1: Design points and elicited probabilities

age	famhist	tobacco	Expert1_Median	Expert2_25%	Expert2_75%
20	Absent	0.00	0.05	0.05	0.075
29	Absent	1.50	0.11	0.08	0.13
32	Absent	0.00	0.1	0.08	0.125
34	Absent	0.40	0.12	0.10	0.13
40	Absent	1.62	0.165	0.14	0.18
41	Present	0.90	0.33	0.32	0.39
42	Absent	2.92	0.2	0.18	0.22
43	Absent	6.75	0.275	0.25	0.31
46	Absent	12.00	0.46	0.39	0.50
54	Present	7.60	0.68	0.65	0.74
55	Present	4.80	0.64	0.59	0.68
61	Present	25.01	0.96	0.93	0.98
64	Present	8.60	0.8	0.77	0.83
64	Present	12.18	0.86	0.82	0.9

References

- Choy, S.L., O'Leary, R. and Mengersen, K., 2009. Elicitation by design in ecology: using expert opinion to inform priors for Bayesian statistical models. Ecology, 90(1), 265-277.
- Hosack, G.R., Hayes, K.R. and Barry, S.C., 2017. Prior elicitation for Bayesian generalised linear models with application to risk control option assessment. Reliability Engineering & System Safety, 167: 351-361.
- Rousseauw, J., du Plessis, J., Benade, A., Jordaan, P., Kotze, J. and Ferreira, J. (1983). Coronary risk factor screening in three rural communities, South African Medical Journal 64: 430-436.